

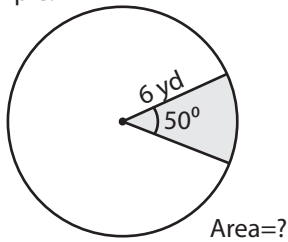
Name : _____

Score : _____

Area of a Sector

Sheet 1

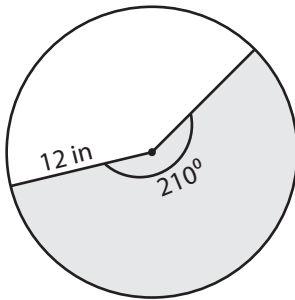
Example:



$$\begin{aligned}\text{Area of a sector} &= \frac{\text{central angle}}{360^\circ} \times \pi \times \text{radius}^2 = \frac{\theta \times \pi \times r^2}{360^\circ} \\ &= \frac{50^\circ \times 3.14 \times 6 \times 6}{360^\circ} \\ &= \mathbf{15.7 \text{ yd}^2}\end{aligned}$$

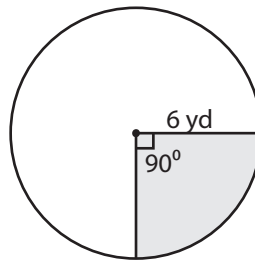
Find the area of each shaded region. Round the answer to two decimal places. (use $\pi=3.14$)

1)



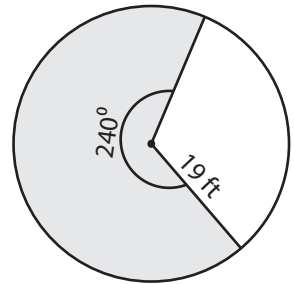
Area = _____

2)



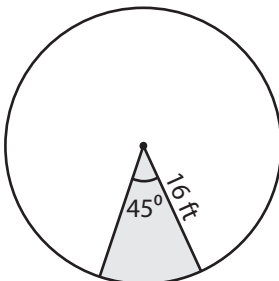
Area = _____

3)



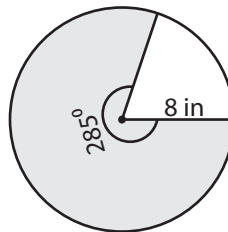
Area = _____

4)



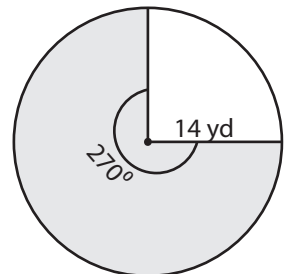
Area = _____

5)



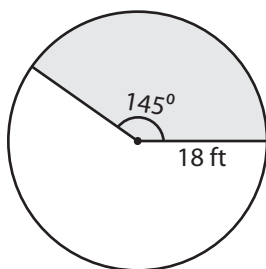
Area = _____

6)



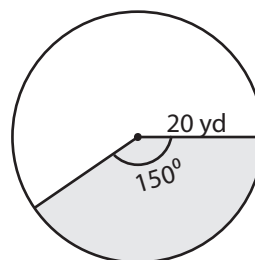
Area = _____

7)



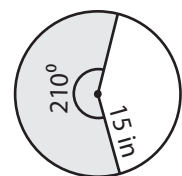
Area = _____

8)



Area = _____

9)



Area = _____

Name : _____

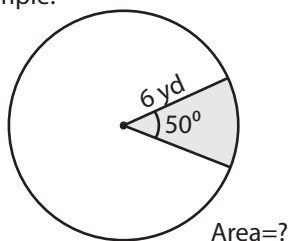
Answer Key

Score : _____

Area of a Sector

Sheet 1

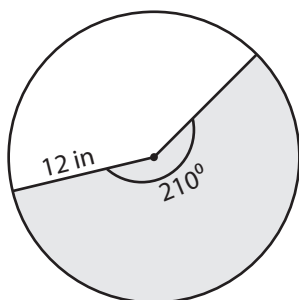
Example:



$$\begin{aligned}\text{Area of a sector} &= \frac{\text{central angle}}{360^\circ} \times \pi \times \text{radius}^2 = \frac{\theta \times \pi \times r^2}{360^\circ} \\ &= \frac{50^\circ \times 3.14 \times 6 \times 6}{360^\circ} \\ &= \mathbf{15.7 \text{ yd}^2}\end{aligned}$$

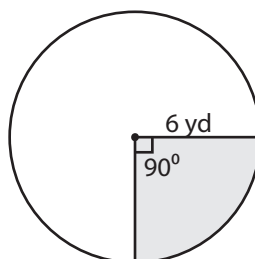
Find the area of each shaded region. Round the answer to two decimal places. (use $\pi=3.14$)

1)



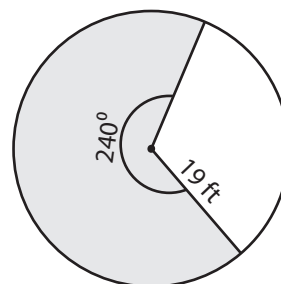
Area = **263.76 in²**

2)



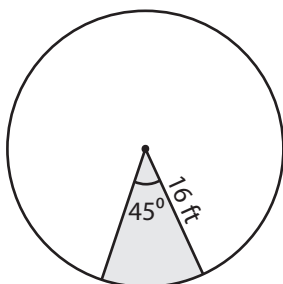
Area = **28.26 yd²**

3)



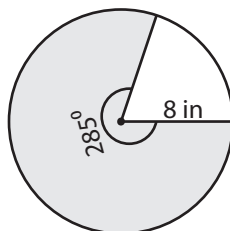
Area = **755.69 ft²**

4)



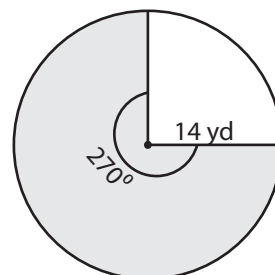
Area = **100.48 ft²**

5)



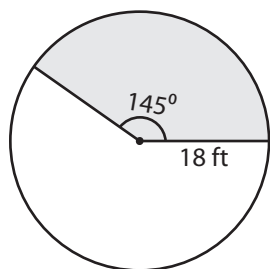
Area = **159.09 in²**

6)



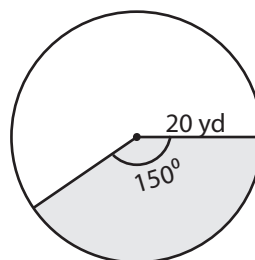
Area = **461.58 yd²**

7)



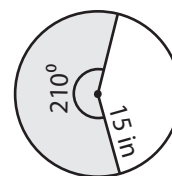
Area = **409.77 ft²**

8)



Area = **523.33 yd²**

9)



Area = **412.13 in²**